Resources and Development

The big idea

Everything in our environment that can be used to satisfy our needs, provided it is **technologically accessible, economically feasible and culturally acceptable**, is a **resource**. Resources are not free gifts of nature: they become resources only through **human activities**. Because humans have used them greedily, we need planning and conservation, so that development is **sustainable**.

Types of resources

Basis	Types	Examples
Origin	Biotic (from the biosphere, have life) / Abiotic (non-living)	Humans, flora, fauna, fisheries / rocks, metals
Exhaustibility	Renewable (can be renewed or reproduced) / Non-renewable (take millions of years)	Solar, wind, water, forests / fossil fuels, minerals
Ownership	Individual, Community, National, International	Plots, houses / village ponds, grazing grounds / roads, canals, minerals / oceanic resources beyond 200 nautical miles
Status of development	Potential, Developed, Stock, Reserves	see below

- **Potential resources**: found in a region but not yet utilised (e.g., wind and solar energy in Rajasthan and Gujarat).
- **Developed resources**: surveyed, and their quality and quantity determined for use.
- **Stock**: materials that can satisfy human needs but we lack the appropriate technology to use them (e.g., using **hydrogen** from water as energy).
- **Reserves**: a subset of stock that can be used with existing technology but whose use has not yet started (e.g., water in dams kept for the future).

EXCLUSIVE ECONOMIC ZONE (EEZ)

India has the right to exploit resources in the ocean up to **200 nautical miles (about 370 km)** from its coast. Oceanic resources beyond this belong to **international institutions**.

Development of resources and sustainable development

Indiscriminate use of resources has caused (1) **depletion** of resources to satisfy the greed of a few, (2) **accumulation** of resources in a few hands, dividing society into rich and poor, and (3) **global ecological crises** like global warming, ozone layer depletion, pollution and land degradation.

SUSTAINABLE DEVELOPMENT

Development should take place **without damaging the environment**, and development in the present should **not compromise the needs of future generations**.

Rio de Janeiro Earth Summit (June 1992): over 100 heads of state met in Brazil to address environmental protection and socio-economic development. They signed the **Declaration on Global Climatic Change and Biological Diversity** and adopted **Agenda 21**, which aims at achieving global sustainable development,

combating environmental damage, poverty and disease through global cooperation. Every local government should draw its own local Agenda 21.

Other milestones: the **Club of Rome** advocated resource conservation in **1968**; **Schumacher's** book *Small is Beautiful* (1974) presented Gandhian philosophy; the **Brundtland Commission Report (1987)**, *Our Common Future*, introduced the concept of sustainable development. Gandhiji said: "**There is enough for everybody's need and not for anybody's greed.**"

Resource planning

India has enormous diversity: some regions are rich in some resources but poor in others. **Jharkhand, Madhya Pradesh and Chhattisgarh** are rich in minerals and coal; **Arunachal Pradesh** has abundant water but lacks infrastructure; **Rajasthan** is rich in solar and wind energy but lacks water; **Ladakh** has a rich cultural heritage but is deficient in water, infrastructure and some vital minerals.

Steps of resource planning:

1. **Identification and inventory** of resources across regions (surveying, mapping, qualitative and quantitative estimation).
2. **Evolving a planning structure** with appropriate technology, skill and institutions.
3. **Matching resource development plans with overall national development plans.**

India has made efforts towards resource planning since the **First Five Year Plan**. Colonies were exploited because colonisers had better technology, which shows that resources contribute to development only with technology and institutions.

Land resources and land use

India's land under different relief features: about **43% plains** (agriculture and industry), **30% mountains** (perennial rivers, tourism, ecology) and **27% plateaus**

(minerals, fossil fuels, forests).

Land use pattern. India's total geographical area is **3.28 million sq km**, but land use data is available for only about **93%** (most of the north-east except Assam, and parts of Jammu and Kashmir occupied by Pakistan and China, have not been surveyed). Land use categories: forests; land not available for cultivation (barren and wasteland; non-agricultural uses); other uncultivated land (permanent pastures, culturable waste, tree crops); fallow lands (current fallow, left uncultivated for less than a year; other fallow, 1 to 5 years); and **net sown area**.

FACTS EXAMINERS LIKE

- Net sown area is over **80%** of total area in **Punjab and Haryana**, and less than **10%** in **Arunachal Pradesh, Mizoram, Manipur and the Andaman and Nicobar Islands**.
- The **National Forest Policy (1952)** outlined a desired **33%** of geographical area under forest. Actual forest area is far below this.
- **Wasteland** includes rocky, arid and desert areas, and land put to non-agricultural uses (settlements, roads, railways, industry).

Land degradation. About **130 million hectares** of land are degraded in India (NCERT figure). Causes:

- **Mining**: in **Jharkhand, Chhattisgarh, Madhya Pradesh and Odisha**, deforestation due to mining and abandoned deep scars.
- **Overgrazing**: in **Gujarat, Rajasthan, Madhya Pradesh and Maharashtra**.
- **Over-irrigation**: in **Punjab, Haryana and western Uttar Pradesh**, causing waterlogging and increase in **salinity and alkalinity**.
- **Mineral processing**: grinding of limestone for cement and calcite and soapstone for ceramics generates huge dust that retards infiltration of water into soil.
- **Industrial effluents** as waste.

Conservation measures: afforestation and proper management of grazing; planting **shelter belts**; control of overgrazing; stabilisation of sand dunes by

growing thorny bushes; proper management of wastelands; control of mining; proper discharge of industrial effluents after treatment.

Soil types and conservation

Soil is the most important renewable natural resource. It takes **millions of years** to form soil up to a few centimetres deep. Factors: relief, parent rock, climate, vegetation, other life forms and time; also changes in temperature, running water, wind, glaciers, decomposers.

MAJOR SOILS OF INDIA (WITH MAP AREAS)

1. Alluvial soil: the most widely spread and important soil; the entire **northern plains** (deposited by the Indus, Ganga and Brahmaputra), also **Rajasthan and Gujarat** through a narrow corridor, and the **eastern coastal plains**, especially the deltas of the **Mahanadi, Godavari, Krishna and Kaveri**. Consists of sand, silt and clay. By age: **Bangar** (old alluvial, more **kanker** nodules) and **Khadar** (new alluvial, more fertile, finer particles). Very fertile; rich in **potash, phosphoric acid and lime**; ideal for sugarcane, paddy, wheat and other cereals and pulses. Intensively cultivated and densely populated.

2. Black soil (regur): made of extremely fine clayey material, formed from **lava rocks (Deccan trap)**. Covers the plateaus of **Maharashtra, Saurashtra, Malwa, Madhya Pradesh and Chhattisgarh**, extending south-east along the **Godavari and Krishna** valleys. Ideal for **cotton** (also called **black cotton soil**). Well known for its **capacity to hold moisture**. Rich in **calcium carbonate, magnesium, potash and lime**, but **poor in phosphoric contents**. Develops **deep cracks** in hot weather, which helps aeration; sticky when wet.

3. Red and yellow soil: develops on **crystalline igneous rocks** in areas of low rainfall in the **eastern and southern parts of the Deccan plateau**; also parts of **Odisha, Chhattisgarh**, southern parts of the middle Ganga plain and along the piedmont zone of the Western Ghats. **Reddish** due to diffusion of **iron** in crystalline and metamorphic rocks; looks **yellow**

when hydrated.

4. Laterite soil: from the Latin *later* (**brick**). Develops under **high temperature and heavy rainfall**, resulting from intense **leaching**. Humus content low (bacteria destroy it in high temperature). Found in **Karnataka, Kerala, Tamil Nadu, Madhya Pradesh** and hilly areas of **Odisha and Assam**. With manures and fertilisers, suitable for **tea and coffee** (Karnataka, Kerala, Tamil Nadu); red laterite soils in Tamil Nadu, Andhra Pradesh and Kerala suit **cashew nut**.

5. Arid soil: red to brown, generally **sandy** and **saline**; in some areas salt is obtained by evaporating water. Lower horizons have **kankar** due to increasing calcium content downwards, which restricts infiltration of water. Found in **western Rajasthan**; can be cultivated after proper irrigation.

6. Forest soil: in hilly and mountainous areas with sufficient rain forests. Loamy and silty in valley sides, coarse-grained in upper slopes. Acidic with low humus in snow-covered Himalayan areas; fertile in lower valley parts.

Soil erosion is the denudation of the soil cover and subsequent washing down. Causes: deforestation, overgrazing, construction and mining, defective farming methods (wrong ploughing), and natural forces (wind, glacier, water).

- **Gullies**: running water cuts through clayey soils making deep channels. Land becomes unfit for cultivation and is called **bad land**; in the **Chambal basin** such lands are called **ravines**.
- **Sheet erosion**: water flows as a sheet over large areas down a slope, washing away the topsoil.
- **Wind erosion**: wind blows loose soil off flat or sloping land.

SOIL CONSERVATION METHODS

- **Contour ploughing**: ploughing along contour lines decelerates the flow of water down slopes.
- **Terrace cultivation**: steps cut out on slopes restrict erosion; practised in the **western and central Himalayas**.

- **Strip cropping**: large fields divided into strips, with strips of grass left between crops to break the force of wind.
- **Shelter belts**: planting lines of trees to create shelter; contributed to stabilisation of sand dunes and the desert in **western India**.
- Plugging gullies and making checks.

BLACK SOIL AND PHOSPHORUS

Black soil is rich in calcium carbonate, magnesium, potash and lime, but **poor** in phosphoric content. Alluvial soil is the one rich in phosphoric acid.

BOARD FAVOURITES AND MAP WORK

"Classify resources on the basis of ownership/status of development", "Three stages of resource planning", "Features of alluvial/black/laterite soil" (5 marks), "Methods of soil conservation", "Causes of land degradation". **Map:** be able to identify the **major soil types** on an outline map of India (alluvial in northern plains and deltas, black on the Deccan lava plateau, red and yellow in east and south Deccan, laterite in Western Ghats/Kerala, arid in western Rajasthan).

Quick recap

- Resources classified by origin, exhaustibility, ownership and status of development.
- Sustainable development; Rio Earth Summit 1992 and Agenda 21; Brundtland Report 1987.
- Resource planning: identify and inventory, planning structure, match with national plans.
- Land: 43% plains, 30% mountains, 27% plateaus; forest target 33% (1952 policy).

- Six soil types; black soil holds moisture and suits cotton; laterite from leaching.
- Conservation: contour ploughing, terracing, strip cropping, shelter belts.

Practice questions

1. Using hydrogen from water as a source of energy is an example of which type of resource?
   (a) Potential (b) Developed (c) Stock (d) Reserve
2. Agenda 21 was signed at the
   (a) Club of Rome meeting, 1968 (b) Rio de Janeiro Earth Summit, 1992 (c) Brundtland Commission, 1987 (d) Kyoto Conference, 1997
3. Which soil is also known as regur and is ideal for growing cotton?
   (a) Alluvial soil (b) Laterite soil (c) Arid soil (d) Black soil
4. Over-irrigation is the main cause of land degradation in
   (a) Punjab, Haryana and western Uttar Pradesh (b) Jharkhand and Odisha (c) Gujarat and Rajasthan (d) Kerala and Karnataka
5. Laterite soil develops mainly due to
   (a) deposition by rivers (b) weathering of lava rocks (c) intense leaching under high temperature and heavy rainfall (d) wind deposition in deserts
6. Terrace cultivation as a soil conservation method is practised mostly in
   (a) the Thar desert (b) the Ganga delta (c) the Malwa plateau (d) the western and central Himalayas
7. The National Forest Policy (1952) set a desired forest cover of
   (a) 20 per cent (b) 33 per cent (c) 43 per cent (d) 50 per cent
8. Distinguish between bangar and khadar.
9. What is resource planning? Explain the three stages of resource planning in India.

Answers: 1. c 2. b 3. d 4. a 5. c 6. d 7. b 8. see notes 9. see notes